

ISLAMIC UNIVERSITY OF TECHNOLOGY



VISUAL PROGRAMMING LAB

CSE 4402

Getting started with JavaFX & SceneBuilder

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1 Introduction to JavaFX & SceneBuilder

JavaFX is an open source, next generation client application platform for desktop, mobile and embedded systems built on Java. It is a collaborative effort by many individuals and companies with the goal of producing a modern, efficient, and fully featured toolkit for developing rich client applications. JavaFX allows you to create Java applications with a modern, hardware-accelerated user interface that is highly portable.

Scene Builder is a visual layout tool primarily used for designing user interfaces (UIs) for JavaFX applications. It allows users to create UIs by dragging and dropping UI components into a design area, without needing to write any code. The tool automatically generates the FXML code for the UI layout as it is created.

2 Getting Started with JavaFX

You may follow this [tutorial](#) for installing JavaFX. The steps to install JavaFX:

- Go to this website: <https://gluonhq.com/products/javafx/> and download the appropriate SDK for your operating system.
- Unzip the file and keep track of where you kept it.
- Open the Eclipse IDE and got to: Help → Eclipse Marketplace → Install **e(fx) eclipse**
- Create a new JavaFX project
- Create a user defined library following: Window → Preferences → Java → Build Path → User Libraries → New → Name it JavaFX → Add External JARs → Select the extracted jar files in the JavaFX **lib** folder
- Set the user library you just created by: Right click on your project → Buildpath → Add Libraries → User Library → Add JavaFX
- Configure the build path and keep all your libraries under **ModulePath**.

- Add a Virtual Machine Argument by: Right click on your project → Run As → Run Configurations → The arguments tab → VM arguments → paste: `--module-path "path_to_javafx-sdk_lib" --add-modules javafx.controls,javafx.fxml`
- Press run to finish the setup

3 Getting Started with SceneBuilder

You may follow this [tutorial](#) for installing SceneBuilder. The steps to install SceneBuilder:

- Go to this website: <https://gluonhq.com/products/scene-builder/> and download the appropriate installer for your operating system.
- Install SceneBuilder and keep track of where you kept the `SceneBuilder.exe`.
- Create a new FXML file.
- To link the SceneBuilder executable: Window → Preferences → JavaFX → Browse for the location where you kept the `SceneBuilder.exe`.
- Right click on the FXML file → Open with SceneBuilder
- To link your code with the FXML file, add:

```
Parent root = FXMLLoader.load(getClass().getResource("
    File_Name.fxml"));
```